



Chi-Squared Tests · Worksheet 1

Further Statistics 1 Chapter 6: Chi-Squared Tests. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Use $X^2 = \sum \frac{(O - E)^2}{E}$, and pool classes until every expected frequency is at least 5. Critical values are given in each question. In every test, state your hypotheses, the degrees of freedom and a conclusion **in context**.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Goodness-of-fit tests (given proportions)	20	
B Goodness-of-fit tests for binomial and Poisson models	20	
C Tests for association (contingency tables)	22	
Roar-level challenge	5	
Total	67	

Section A Goodness-of-fit tests (given proportions)

20 marks

1. State the number of degrees of freedom for each goodness-of-fit test. [2]
- (a) Testing whether a six-sided die is fair, using the frequencies of the six scores.
- (b) Testing whether a Poisson model fits data, where the mean is estimated from the data and there are 5 classes after pooling.

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2. A spinner has four sectors, A , B , C and D . It is spun 80 times, with the results below.

[3]

Sector	A	B	C	D
Frequency	25	18	22	15

Assuming the spinner is fair, write down the expected frequencies and calculate the value of the test statistic X^2 .

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3. A die is rolled 120 times, with the results below.

[3]

Score	1	2	3	4	5	6
Frequency	15	22	18	26	14	25

Test, at the 5% level of significance, whether the die is fair. [$\chi^2_5(5\%) = 11.070$]

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4. Genetic theory predicts that four types of plant, P , Q , R and S , occur in the ratio 9 : 3 : 3 : 1. In a sample of 160 plants, the frequencies are shown below.

[4]

Type	P	Q	R	S
Frequency	98	25	27	10

Test, at the 5% level of significance, whether the data are consistent with the theory. [$\chi^2_3(5\%) = 7.815$]

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5. The number of accidents in a factory on each day of the working week over one year is recorded.

Day	Mon	Tue	Wed	Thu	Fri
Accidents	30	18	20	17	35

The manager wants to test whether accidents are equally likely to occur on each day.

(a) State suitable hypotheses and write down the expected frequencies under the null hypothesis. [2]

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(b) Carry out the test at the 5% level of significance. [$\chi^2_4(5\%) = 9.488$] [4]

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(c) State, with a reason, whether your conclusion would be the same at the 1% level of significance, and comment on the days with the largest contributions to X^2 . [$\chi^2_4(1\%) = 13.277$] [2]

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(Total for Question 5 is 8 marks)



Section B Goodness-of-fit tests for binomial and Poisson models

20 marks

6. A Poisson model is fitted to a set of data grouped into 6 classes. No classes need to be pooled. State the number of degrees of freedom for the goodness-of-fit test if [2]
- (a) the mean is given as 2,
- (b) the mean is estimated from the data.

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7. The number of girls in each of 50 families with three children is recorded. [3]

Number of girls	0	1	2	3
Number of families	5	17	20	8

A binomial model $B(3, p)$ is to be fitted. Estimate p , and find the expected number of families with exactly 2 girls.

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8. Explain why, in a goodness-of-fit test, the number of degrees of freedom is reduced by one for each parameter estimated from the data, and why classes with expected frequencies below 5 are pooled. [3]

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9. Seeds are sold in packets of 4. The supplier claims that each seed germinates with probability 0.7, independently. The numbers of seeds germinating in 150 packets, and the expected frequencies under $B(4, 0.7)$, are shown below. [4]

Number germinating	0	1	2	3	4
Observed	3	15	42	55	35
Expected	1.22	11.34	39.69	61.74	36.01

Test, at the 5% level of significance, whether the supplier's claim is supported. [$\chi^2_3(5\%) = 7.815$]

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10. The number of calls, X , received by a helpline in each of 80 randomly chosen 5-minute intervals is recorded.

x	0	1	2	3	4	5
Frequency	14	18	20	16	6	6

There were no intervals with more than 5 calls. A Poisson model is to be fitted.

- (a) Show that the mean number of calls per interval is 2. [1]

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- (b) Using $Po(2)$, some of the expected frequencies are given below. Find the values of a and b . [2]

x	0	1	2	3	4	≥ 5
Expected	10.83	21.65	21.65	a	7.22	b

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- (c) Test, at the 5% level of significance, whether a Poisson distribution is a suitable model. [5]
[$\chi^2_3(5\%) = 7.815$]

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(Total for Question 10 is 8 marks)

**Section C Tests for association (contingency tables)**

22 marks

11. A contingency table has 3 rows and 4 columns. One cell has row total 60 and column total 45, and the grand total is 200. [2]
State the number of degrees of freedom for a test of association, and find the expected frequency for this cell.

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12. 100 sixth-form students are asked which type of subject they prefer. [3]

	Maths	Science	Arts	Total
Year 12	20	25	15	60
Year 13	20	15	5	40
Total	40	40	20	100

Find the expected frequencies for a test of whether preferred subject is independent of year group.

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13. The grades obtained by students at two schools are shown below. [3]

	Distinction	Merit	Pass
School A	12	18	30
School B	18	12	10

The expected frequencies are 18, 18, 24 for school A and 12, 12, 16 for school B.

Test, at the 5% level, whether there is an association between school and grade. [$\chi^2_{2}(5\%) = 5.991$]

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14. 130 people are asked how they travel to work. The expected frequencies for a test of association between age group and method of travel are shown below. [4]

Expected	Bus	Car	Cycle
Under 30	17.31	23.08	9.62
30 to 60	24.23	32.31	13.46
Over 60	3.46	4.62	1.92

(a) Explain why the rows “30 to 60” and “Over 60” should be combined, and state the degrees of freedom after combining.

(b) After combining, $X^2 = 10.17$. Complete the test at the 5% level. [$\chi^2_2(5\%) = 5.991$]

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15. 200 people are asked which type of film they prefer. The results are shown below.

	Action	Comedy	Drama	Total
16–25	30	25	10	65
26–40	20	30	20	70
Over 40	10	25	30	65
Total	60	80	60	200

The values of $\frac{(O - E)^2}{E}$ for some cells, to 4 decimal places, are shown below.

$\frac{(O - E)^2}{E}$	Action	Comedy	Drama
16–25	p	0.0385	4.6282
26–40	0.0476	0.1429	0.0476
Over 40	q	0.0385	5.6538

(a) State suitable hypotheses for a test of association. [1]

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(b) Show that the expected frequency for 26–40 and Comedy is 28, and find the expected frequency for 16–25 and Action. [2]

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(c) Find the values of p and q . [2]

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(d) Test, at the 1% level of significance, whether there is an association between age group and preferred type of film. [$\chi^2_4(1\%) = 13.277$] [4]

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(e) Use the table to describe the association. [1]

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(Total for Question 15 is 10 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A 2×2 contingency table has observed frequencies a, b in the first row and c, d in the second row, with grand total $n = a + b + c + d$. [5]

(a) Show that, for the cell containing a , $O - E = \frac{ad - bc}{n}$, and deduce that $|O - E|$ is the same for all four cells.

(b) Hence find X^2 for the table with first row 30, 20 and second row 10, 40.

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END OF WORKSHEET

Total: 67 marks